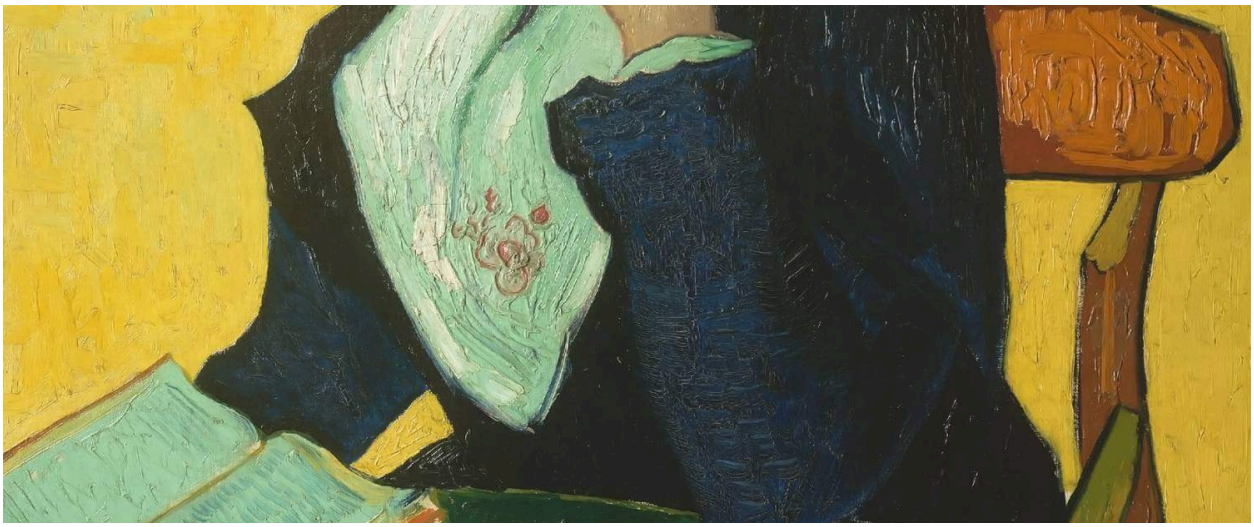

Basics Of Mechatronics Project

Modular Integration of the Three Mechanisms

1. Modules Design and Construction
    a. Shooting Mechanism
        i. Mechanical Design and Prototyping
            1. Drafting Mechanism Design
            2. Motor Type Selection
            3. Inventor Mechanical Design
            4. Mechanism Calculations
            5. Actuator Sizing
            6. Material Selection
            7. Mechanical Parts Fabrication
        ii. Electrical Wiring
            1. **Circuit Design on Proteus**
            2. **Battery Sizing**
            3. Mechanism Current Consumption

        4. Wiring Electronic Components
    iii. Coding and Firmware
        1. Microcontroller Selection
        2. Sensor Selection
        3. Sensors Calibrations
        4. Components Testing
        5. Control Algorithm Design
        6. Mechanism Flowcharts and Block Diagram
        7. Embedded Software Deployment
        8. Overall Mechanism Testing
b. Lifting Mechanism
    i. Mechanical Design and Prototyping
        1. Drafting Mechanism Design
        2. Motor Type Selection
        3. Inventor Mechanical Design
        4. Mechanism Calculations
        5. Actuator Sizing
        6. Material Selection
        7. Mechanical Parts Fabrication
    ii. Electrical Wiring
        1. **Circuit Design on Proteus**
        2. **Battery Sizing**
        3. Mechanism Current Consumption
        4. Wiring Electronic Components
    iii. Coding and Firmware
        1. Microcontroller Selection

2. Sensors Selection
3. Sensor Calibrations
4. Components Testing
5. Control Algorithm Design
6. Mechanism Flowcharts and Block Diagram
7. Embedded Software Deployment
8. Overall Mechanism Testing

c. Sorting Mechanism

i. Mechanical Design and Prototyping

1. Drafting Mechanism Design
2. Motor Type Selection
3. Inventor Mechanical Design
4. Mechanism Calculations
5. Actuator Sizing
6. Material Selection
7. Mechanical Parts Fabrication

ii. Electrical Wiring

1. **Circuit Design on Proteus**
2. **Battery Sizing**
3. Mechanism Current Consumption
4. Wiring Electronic Components

iii. Coding and Firmware

1. Microcontroller Selection
2. Sensors Selection
3. Sensor Calibrations
4. Components Testing

5. Control Algorithm Design
6. Mechanism Flowcharts and Block Diagram
7. Embedded Software Deployment
8. Overall Mechanism Testing

d. Interface
   i. LCD interface
   ii. Processing.io interface
   iii. Bluetooth Integration
   iv. User Input Integration
   v. Inter-MCU Communication
   vi. PCB Schematic

2. Assembly and Integration
   a. Mechanical Assembly
      i. Inter-Mechanism Track Design and Fabrication
      ii. Wire Casing Design and Fabrication
   b. Electrical Integration
      i. Connection Plugs
      ii. Start/Stop Buttons
      iii. Overall Current Consumption
      iv. Distributed Components Testing
      v. Overall Testing
   c. Documentation
      i. Toy Theme Research and Generation
      ii. Toy Idea Research and Generation
      iii. Target Audience Research
      iv. Gantt Charts

v. PDS
vi. Presentation
vii. GIT repository
viii. Filming the Project Video
ix. Technical Report